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Course: Networks System Design

Instructor: KUY Movsun

Assignment: Lab1

Due Date: October 28, 2025 (12:00 AM)

Part 1: Identifying Your Host and Edge Connection

Open the Command Line/Terminal

```
C:\Users\User>ipconfig
```

```
Wireless LAN adapter Wi-Fi 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::1483:838:d66e:4ee4%1
   IPv4 Address. . . . . . . . . . . : 192.168.204.204
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.204.254

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

```
My Host IP Address: 192.168.204.204
My Default Gateway: 192.168.204.254
```

Part 2: Testing Connectivity and Latency

1. ping 127.0.0.1

```
C:\Users\User>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

2. ping Default Gateway: 192.168.204.254

```
C:\Users\User>ping 192.168.204.254

Pinging 192.168.204.254 with 32 bytes of data:
Reply from 192.168.204.254: bytes=32 time=186ms TTL=64
Reply from 192.168.204.254: bytes=32 time=12ms TTL=64
Reply from 192.168.204.254: bytes=32 time=5ms TTL=64
Reply from 192.168.204.254: bytes=32 time=6ms TTL=64

Ping statistics for 192.168.204.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 186ms, Average = 52ms
```

3. ping google.com

```
C:\Users\User>ping google.com

Pinging google.com [172.217.25.110] with 32 bytes of data:
Reply from 172.217.25.110: bytes=32 time=303ms TTL=116
Request timed out.
Reply from 172.217.25.110: bytes=32 time=94ms TTL=116
Reply from 172.217.25.110: bytes=32 time=254ms TTL=116

Ping statistics for 172.217.25.110:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 94ms, Maximum = 303ms, Average = 217ms
```

Table

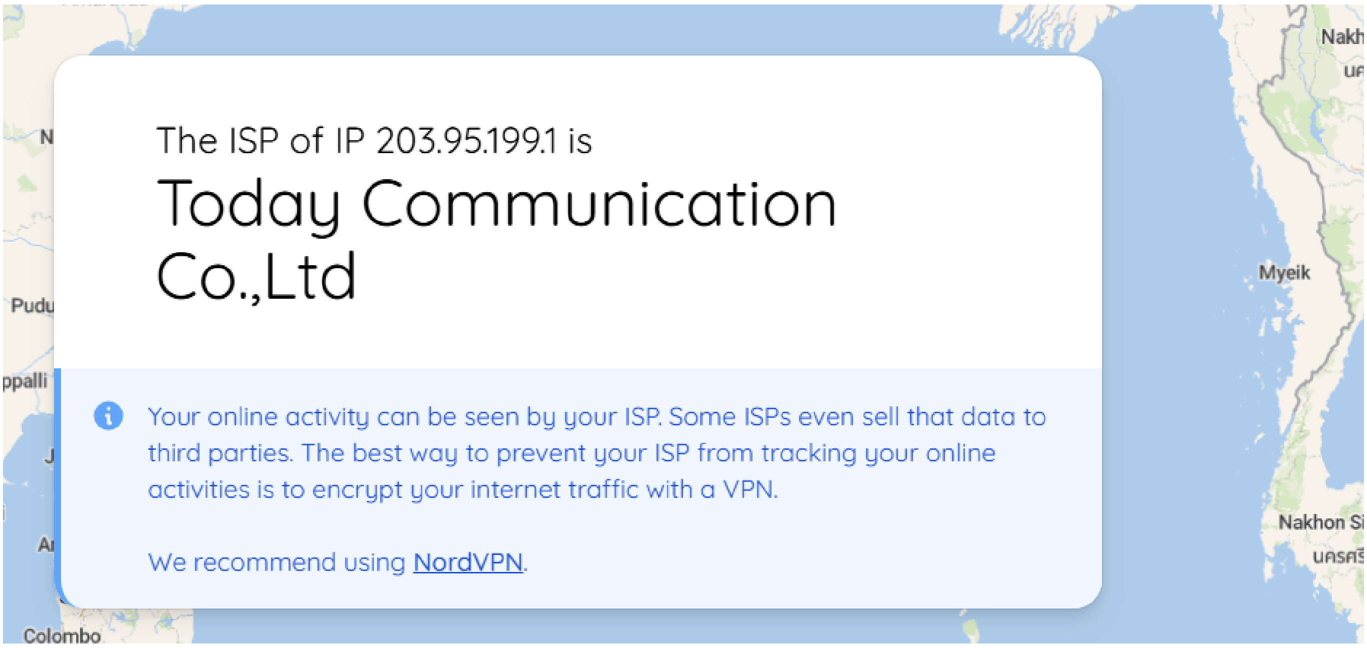
Test Target	Min RTT (ms)	Max RTT (ms)	Avg RTT (ms)	Packet Loss (%)
127.0.0.1	0	0	0	0%
192.168.204.254	5	186	52	0%
google.com	94	303	217	25%

Part 3: Tracing the Path Through the Network Core

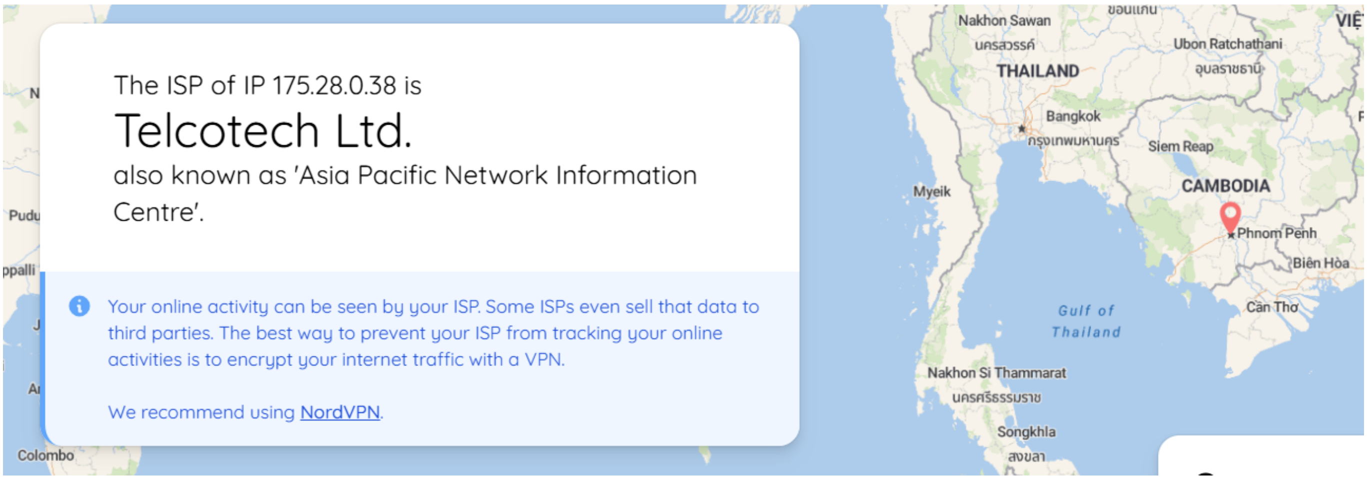
Execute the Trace Command (tracert google.com) and Identify Core Components: List the IP address and RTT for **Hops 2, 5, and 8** (or the closest available public IPs).

```
C:\Users\User>tracert google.com

Tracing route to google.com [142.250.193.238]
over a maximum of 30 hops:

  1     8 ms     5 ms     5 ms  192.168.204.254
  2   314 ms   298 ms     *     203.95.199.1
  3   305 ms    44 ms    18 ms  10.170.170.45
  4   543 ms   277 ms     *     172.16.101.49
  5     *        *        *     Request timed out.
  6    83 ms   437 ms   272 ms  175.28.0.38
  7   210 ms   307 ms   301 ms  192.178.99.69
  8    33 ms   233 ms   279 ms  216.239.48.239
  9    96 ms   295 ms   297 ms  del11s18-in-f14.1e100.net [142.250.193.238]

Trace complete.
```

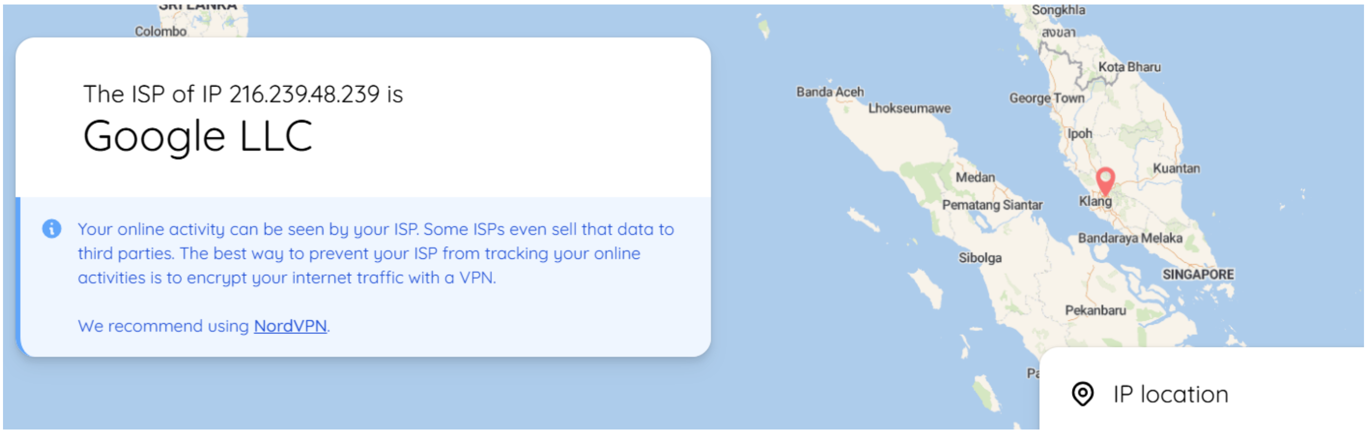
Hop 2:



Hop6:



Hop8:



Hop	IP Address	RTT (ms)	ISP / Organization Name
2	203.95.199.1	298–314	Today Communication Co., Ltd
6	175.28.0.38	83–437	Telcotech Ltd
8	216.239.48.239	33–279	Google LLC

Part 4: Calculating Transmission Delay

Given:

- Packet size (L = 12,000 Kbits)
- Transmission rate (R = 100,000,000 bits/sec)
- Number of links = 5

Calculations:

```
dtrans = L/R =12,000/100,000,000 = 0.00012sec = 0.12 ms
```

```
Total Delay = 5 \* 0.12 = 0.6 ms
```

Submission Answer

Store-and-Forward means each router must receive the entire packet before forwarding it to the next link. This adds delay at every hop, so the total end-to-end delay is the transmission delay multiplied by the number of links.

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Note: Viewing in VsCode IDE for better formatting!!!